

GemEx OTC Desk: Phase 1 MVP Backend API Specification

This document outlines the data models, user roles, API architecture, and endpoint specifications for the Crypto-to-Naira Payout MVP, built using Bun and the Hono Web Framework.

1. Core Technology and Structure

Component	Technology	Role / Dependency
Runtime	Bun	High-performance JavaScript/TypeScript runtime.
Web Framework	Hono	Fast, lightweight web framework used for routing and middleware.
Database	MongoDB	Primary data store for Users, Transactions, Quotes, and Audit Logs.
Validation	Zod (via hono/validator)	Schema definition and request validation.
Module Path	src/modules/[module_name]	Modular structure for features.

2. Core Data Models (MongoDB Schemas)

The following MongoDB schemas represent the core entities managed by the API.

2.1. User Model (User)

Stores client, dealer, and admin information.

Field	Type	Description
_id	ObjectId	Unique user identifier.
email	String	User email (unique).
passwordHash	String	Hashed password.
role	String	(ENUM: 'Client', 'Dealer', 'Admin').
status	String	(ENUM: 'PendingKYC', 'Verified', 'Suspended').
is2FAEnabled	Boolean	Whether 2FA is active (required for staff).
kycDetails	Object	Stores QoreID results (ID, address verification status, risk score).
bankAccount	Object	Verified Naira payout account details (bank name, number, beneficiary). Must be encrypted.

tradeLimits	Object	Configurable daily/weekly/monthly volume limits.
createdAt	Date	Record creation timestamp.
updatedAt	Date	Last update timestamp.

2.2. Transaction Model (Transaction)

Tracks the full lifecycle of a Crypto-to-Naira trade.

Field	Type	Description
_id	ObjectId	Unique transaction identifier.
userId	ObjectId	Reference to the Client who initiated the trade.
quoteId	ObjectId	Reference to the locked Quote used for the trade.
type	String	(ENUM: 'CRYPTO_TO_NAIRA') for MVP.
cryptoAmount	Number	Amount of crypto sent by the client.
cryptoAsset	String	(ENUM: 'USDT', 'BTC', 'USDC', etc.).
cryptoNetwork	String	(ENUM: 'TRC20', 'BSC', 'ERC20', 'BTC').
nairaPayoutAmount	Number	Calculated Naira amount to be paid out.
status	String	(ENUM: 'QuotePending', 'AwaitingDeposit', 'CryptoConfirmed', 'PayoutPending', 'Settled', 'Failed').
depositAddress	String	Unique Tatum-generated deposit address.
cryptoTxId	String	Blockchain transaction ID for the client deposit.
fiatTxRef	String	Moniefty (Moniepoint) transfer reference ID.
timestamps	Object	Detailed timestamps (e.g., depositConfirmedAt , payoutSentAt).

2.3. Quote Model (Quote)

Stores dealer-generated quotes.

Field	Type	Description
_id	ObjectId	Unique quote identifier.
dealerId	ObjectId	Reference to the Dealer who generated the quote.

baseAsset	String	e.g., 'USDT' (Client Sells).
quoteAsset	String	e.g., 'NGN' (Client Receives).
rate	Number	The firm, set NGN rate per unit of crypto (no markup applied).
expiration	Date	Time when the quote becomes invalid.
status	String	(ENUM: 'Active', 'Used', 'Expired', 'Canceled').

2.4. Pricing Configuration Model (PricingConfig)

Stores the current global, indicative rates and platform-wide fees. This rate is set by the Admin and is used as the **System Conversion Rate** for initial Client quote requests.

Field	Type	Description
_id	ObjectId	Unique configuration identifier (only one document expected).
cryptoAsset	String	(ENUM: 'USDT', 'BTC', 'USDC', etc.).
systemRate	Number	The global, indicative NGN rate per unit of crypto used for initial client quote requests.
feePercentage	Number	Optional platform fee applied to the system rate (e.g., 0.5 for 0.5%).
lastUpdatedBy	ObjectId	Reference to the Admin user who last set the rate.
updatedAt	Date	Last update timestamp.

3. Account Types & Privileges

The API defines three primary account roles with distinct permissions for accessing modules and endpoints.

Role	Description	Core Privileges (API Access)
Client	Verified users who initiate trade requests and receive payouts. They are shown the System Conversion Rate for an initial estimate, but require a Dealer-generated Firm Quote Rate to transact.	Auth, KYC/Profile, Quote Request, Transaction History, Notifications.
Dealer	Administrative staff who manually set the firm Naira conversion rate for specific OTC quotes. They monitor deposits, create and manage quotes, and approve manual trades, essentially defining the rate amount per transaction.	All Client access + Quote Generation, Transaction Management, Audit Log View, Payout Monitoring.

Admin	Superuser with full configuration and operational control, including setting the Global System Conversion Rate .	All Dealer access + User Management (KYC, Suspension, Limits), Global Configuration, Pricing Configuration , Third-Party API Key Management.
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4. Notifications Module (src/modules/notify)

The notify module handles all outgoing communication and in-app message triggers.

Event Trigger	Recipient	Delivery Method(s)
New Quote Generated	Client	Email, In-App Notification
Quote Accepted/Trade Initiated	Dealer	In-App Notification
Crypto Deposit Confirmed	Client/Dealer	SMS (Twilio), Email, In-App Notification
Naira Payout Success	Client/Dealer	SMS (Twilio), Email, In-App Notification
Account Status Change (KYC/Suspension)	Client	Email, In-App Notification
Admin Action Taken	Admin/Dealer	Audit Log Entry

5. Module Endpoints & Validation Specifications

The following details the required endpoints for the Crypto-to-Naira MVP flow, including the Zod schemas used for validation (auth.validation.ts , quote.validation.ts , etc.).

5.1. Module: auth (Authentication)

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
/api/v1/auth/register	POST	Create a new Client account.	Public	registerSchema : email (string, email), password (string, min 8)
/api/v1/auth/verify-email/:token	GET	Finalize registration and verify email address using the token sent via email.	Public	Path: token (string)
/api/v1/auth/login	POST	Authenticate and receive JWT/session token.	Public	loginSchema : email (string, email), password (string)
/api/v1/auth/2fa/setup	POST	Initialize 2FA (returns QR code data).	Private (All)	None (Requires JWT)
/api/v1/auth/2fa/verify	POST	Verify 2FA code and enable 2FA.	Private (All)	verify2faSchema : code (string, length 6)

/api/v1/auth/profile	GET	Get current user's profile data.	Private (All)	None
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5.2. Module: kyc (KYC & Profile Management)

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
/api/v1/kyc/submit	POST	Submit KYC information to QoreID.	Private (Client)	kycSubmitSchema : idType , idNumber , proofOfAddress (e.g., base64 file)
/api/v1/kyc/bank	POST	Verify and save client's Naira bank account (Moniefy Account Verification).	Private (Client)	bankVerifySchema : accountNumber (string), bankCode (string)
/api/v1/kyc/bank/verify	POST	QoreID Webhook endpoint for Identity Verification status (Not Moniefy).	Public (QoreID Only)	qoreIdKycWebhookSchema (Requires signature verification)

5.3. Module: quote (Trading & Pricing)

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
/api/v1/quotes	POST	Client: Request a trade quote. Returns an Indicative Rate based on the Admin's Global System Rate.	Private (Client)	requestQuoteSchema : cryptoAsset (string), cryptoAmount (number, > 0)
/api/v1/quotes	GET	Dealer/Admin: Get paginated list of all quotes.	Private (Dealer, Admin)	Query: paginationSchema & quoteListQuerySchema (status filter)
/api/v1/quotes/generate	POST	Dealer/Admin: Generate and send a Firm Quote to a client with a manually set rate .	Private (Dealer, Admin)	generateQuoteSchema : userId , cryptoAsset , rate , expirationMinutes (number)

/api/v1/quotes/:id	GET	Get quote details.	Private (Client, Dealer)	Path: id (objectIdSchema)
/api/v1/quotes/:id/accept	POST	Client: Accept the quote, triggering a Transaction.	Private (Client)	Path: id (objectIdSchema)

5.4. Module: trade (Transaction Flow & Settlement)

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
/api/v1/trades	GET	Get list of user's transactions/trades. (Paginated)	Private (All)	Query: paginationSchema & tradeListQuerySchema (status (string, optional))
/api/v1/trades/:id	GET	Get transaction details (Client/Dealer view).	Private (All)	Path: id (objectIdSchema)
/api/v1/trades/:id/payout	POST	Dealer: Manually trigger Naira payout.	Private (Dealer)	Path: id (objectIdSchema)

5.5. Module: wallet (Crypto Inflow Monitoring)

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
/api/v1/wallet/webhook	POST	Tatum.io Webhook for deposit confirmation. (Monitors assigned deposit addresses)	Public (Tatum Only)	tatumDepositWebhookSchema (Requires signature verification)
/api/v1/wallet/sweep	POST	Admin: Manually trigger hot wallet sweep to cold storage.	Private (Admin)	None (Requires JWT)

5.6. Module: fiat (Naira Payout)

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
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/api/v1/fiat/webhook	POST	Moniefy Webhook for Naira transfer status. (Confirms final payout success/failure.)	Public (Moniefy Only)	moniefyTransferWebhookSchema (Requires signature verification)
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5.7. Module: admin (Staff Management & Configuration)

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
/api/v1/admin/pricing/config	POST	Admin: Set or update the global System Conversion Rate and fees for a crypto asset.	Private (Admin)	systemRateSchema : cryptoAsset (string), systemRate (number), feePercentage (number)
/api/v1/admin/pricing/config	GET	Admin: Get the current global system rates.	Private (Admin)	None
/api/v1/admin/users	GET	Admin: Get paginated list of all Clients.	Private (Admin)	Query: paginationSchema & userListQuerySchema (e.g., status filter)
/api/v1/admin/users/:id	GET	Admin/Dealer: Get the full profile and settings for a specific client.	Private (Admin, Dealer)	Path: id (objectIdSchema)
/api/v1/admin/staff	GET	Admin: Get paginated list of all Dealers/Admins.	Private (Admin)	Query: paginationSchema & staffListQuerySchema (e.g., role filter)
/api/v1/admin/staff/create	POST	Admin: Create a new Dealer or Admin account.	Private (Admin)	staffCreateSchema : email , password , role (ENUM: 'Dealer', 'Admin')
/api/v1/admin/staff/:id/suspend	PUT	Admin: Suspend a staff account.	Private (Admin)	Path: id (objectIdSchema)
/api/v1/admin/staff/:id/activate	PUT	Admin: Activate a suspended staff account.	Private (Admin)	Path: id (objectIdSchema)

/api/v1/admin/users/:id/limit	PUT	Update a client's trade limits.	Private (Admin)	limitSchema : daily , weekly , monthly (number). Path: id (objectIdSchema)
/api/v1/admin/audit	GET	View all audit logs. (Paginated)	Private (Dealer, Admin)	Query: paginationSchema & auditQuerySchema (role , action (optional filters))

5.7.1. Analytics Endpoints

The following endpoints provide aggregate metrics for the Dealer and Admin dashboards.

Route Path	Method	Purpose	Access	Zod Schema (Request Body/Params)
/api/v1/admin/analytics/dashboard	GET	Get all core platform metrics for a single dashboard view: Total Payouts (NGN), Total Deposits (Crypto), Total Amount Locked (Crypto equivalent of Awaiting Payouts), Pending Payouts (Count), and Payout Balance (fiat account balance).	Private (Dealer, Admin)	None

6. Summary of New/Clarified Schemas Needed for Validation

Schema Name	Location	Purpose
objectIdSchema	Shared Utility	Reusable schema to validate MongoDB Object IDs in path parameters.
paginationSchema	Shared Utility	Reusable pagination parameters: page (number, default 1), limit (number, default 20), sortBy (string, optional).
qoreIdKycWebhookSchema	kyc.validation.ts	Validates the QoreID payload for identity verification status updates.
quoteListQuerySchema	quote.validation.ts	Validates filter query parameters (e.g., status=Active) for quote lists.
tradeListQuerySchema	trade.validation.ts	Validates optional query parameters (e.g., status=Settled) for fetching

tatumDepositWebhookSchema	wallet.validation.ts	Validates the Tatum payload for crypto deposit confirmations.
monieftyTransferWebhookSchema	fiat.validation.ts	Validates the Moniefty payload for Naira transfer status updates (success/failure).
userListQuerySchema	admin.validation.ts	Validates filter query parameters (e.g., status=Verified) for client ..